Executive Summary: This week, the team worked on finishing hardware schematics for the WAD. While finishing the schematics for the hardware, we continued working on the software and specifically the signal processing of the Gunshot Detection feature. While this is getting finished up the parts for the WAD had arrived but due to unfortunate circumstances, not all the parts arrived just yet. The team is still missing the battery, charger, breakout board, and diodes needed for completion of the hardware build. This upcoming week we plan on finishing the part gathering, and beginning building of the WAD. We also plan on finishing the rough outlines of software so that once building is done, the testing phase may begin. This week we also plan on having our CDR on Monday 03/30, in which we will give a much needed update to project sponsors.

Hardware:

Cody C.

Progress:

- 3/25 – Team meeting (~30 Minutes)
  - Gathered some of the hardware components and talked about next steps
- Created Schematics for Microphone in KiCad, labels only (~30 Minutes)
- Worked on CDR Presentation
  - Created Slides and Gathered information for slides (~1 Hour 30 Minutes)
  - Create rough idea of what to say during PDR (~30 Minutes)
  - Practiced CDR presentation individually (~ 10 Minutes)
- 3/29 – Team meeting
  - Final touches on CDR presentation (~30 Minutes)
  - Practice CDR Presentation (~30 Minutes)

Total Hours Worked: ~ 4 Hours 10 Minutes

Future Work/Plan

- Practice CDR Presentation as a team
- 3/30 – CDR
- Wait for all parts to come in

Hardware:

Ryan K.

Progress:

- 3/25 - Team Meeting (~30min)
  - Acquired hardware components and talked future plans
- 3/28 - Worked on CDR Slides (~2hrs)
- 3/29 - Overview/Updates of CDR and Team Practice (~1hr)

Total Hours Worked: ~3hrs 30mins

Future Work/Plan:

- Practice CDR with team before presenting on 3/30
- Gather rest of components (Hopefully by Thursday)
- Prepare for the construction of the Wearable Alert Device

Software:

Alex H.

Progress:

- 3/27 (33 min)

    - Investigated methods of making messages from Wearable Alert Device more readable on nRF Connect for Mobile app
    - The nRF Connect for Mobile app sends the human-readable messages in the live logging interface, but it also sends the messages in their hexadecimal form, which causes a large number of hexadecimal values to appear right before the human-readable message.
    - Unable to find a way on the mobile app to disable the hexadecimal format, best way to monitor human-readable messages is to not use the logging display and look at the values directly from the BLE characteristic.
- 3/28 (8 h 32 min)
    - Implemented C struct for a data packet in the software
    - Installed the DFRobot_MSM261.h software library, which handles the inner I2S functionality of the microphone. Read documentation of how the software library works.
    - Calculated buffer sizes for audio sampling and the 1-second audio buffer. Optimized buffer sizes to avoid using all of the Arduino Nano ESP32's SRAM and to not slow down wireless data transmission.
    - Prepared on CDR slides
        - Updated software diagrams with changes and feedback from PDR presentation
        - Added all details for testing wireless data transmission with nRF Connect for Mobile app
        - Added all details for GPS data collection process

    - Added all details for gunshot detection process (including audio sample collection, audio signal processing, and the spectral analysis for possible gunshots)

Total Hours Worked: (9 h 6 min)

Future Work/Plan:

- Fully write the GPS data collection code and correctly store the timestamp, latitude, longitude, and elevation in the data packet. Test this software once the hardware team members complete circuit wiring for the GPS and timing module.
- Fully write the gunshot detection code and correctly store the associated information in the data packet. Test this software after the GPS software has been proven successful.

Project Manager:

Jonas W.

Progress:

- 3/25 – Team meeting (45 Minutes)
    - Early pick up of parts from spring break and then gave parts to Hardware team
    - Scouted the allotted lab for capstone and possible materials
- 3/26 –Revised the Budget (30 Minutes)
    - Added missing parts for the budget and adjusted quantities to be ordered for missing parts on 3/30
- 3/28 – Worked on CDR Slides (2 Hours)
    - Created and finished tables and charts for CDR as well as script for myself
- 3/29 – Team meeting (1 Hour)
    - Final touches on CDR presentation (~30 Minutes)

- Practice CDR Presentation (~30 Minutes)

Total Hours Worked: ~ 4 Hours 15 Minutes

Future Work/Plan

- Practice CDR Presentation as a team
- 3/30 – CDR
- Re order parts and order missing parts
- Wait for all parts to come in